National University

Of Computer & Emerging Sciences Faisalabad-Chiniot Campus

CS 1005 : Discrete Structures

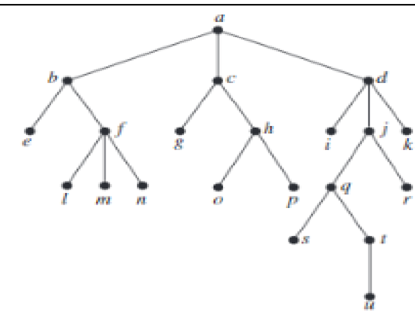
Assignment 6

Instructor: Ms. Mahzaib Younas

Section BCS (C)

Q No 01: Answer these questions about the rooted tree illustrated. [10]

a) Which vertex is the root?
b) Which vertices are internal?
c) Which vertices are leaves?
d) Which vertices are children of j?
e) Which vertex is the parent of h?
f) Which vertices are siblings of o?
g) Which vertices are ancestors of m?
h) Which vertices are descendants of b?



Q No 02: Answer the following and also highlight which formula you have applied [10].

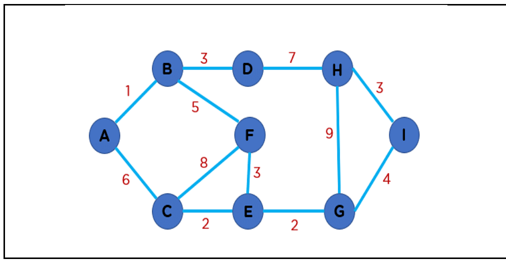
a) How many edges does a tree with 10,000 vertices have?
b) How many vertices does a full 5-ary tree with 100 internal vertices have?
c) How many edges does a full binary tree with 1000 internal vertices have?
d) How many leaves does a full 3-ary tree with 100 vertices have?

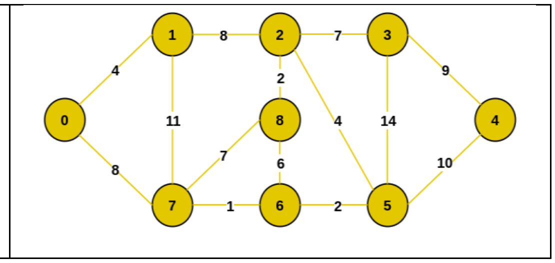
Q No 03: Prove given statements with examples or with tree. A full m-ary tree with **[15]**

(i) N vertices has $i=(n-1)/m$ internal vertices and $l=[(m-1)n+1]/m$ leaves.
(ii) i internal vertices has $n=(mi+1)$ vertices and $l=(m-1)i+1$ leaves.
(iii) L leaves has $n=(ml-1)/(m-1)$ vertices and $i=(l-1)/(m-1)$ internal vertices.

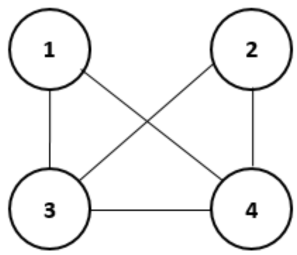
Q No 04: **A chain letter starts when a person sends a letter to five others. Each person who receives the letter either sends it to five other people who have never received it or does not send it to anyone. Suppose that 10,000 people send out the letter before the chain ends and that no one receives more than one letter. How many people receive the letter, and how many do not send it out? Must write formula. [10]**

Q No 05: Use prim's algorithm to find Minimal Spanning tree starting with vertex A of following. [10]





Q No 06: Find number of spanning tree of Graph using Adjacency matrix. [10]

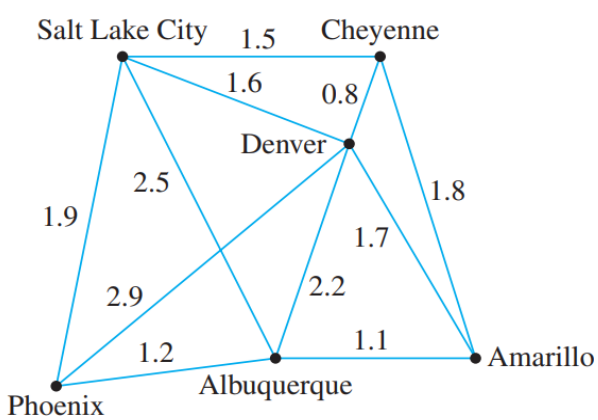


Q No 07: How many different spanning trees does each simple graph has? [15]

a) K_3
b) K_4
c) $K_{2,2}$

Q No 08: Give an example of graph of five vertices and four edges that is not a tree? [5]

Q No 09: A pipeline is to be built that will link six cities. The cost (in hundreds of millions of dollars) of constructing each potential link depends on distance and terrain and is shown in the weighted graph below. Find a system of pipelines to connect all the cities and yet minimize the total cost. [10]



Q No 10: Which of the following graphs are trees? [5]

a) G= (V, E) and V ={a, b, c, d, e}and E={{a, b}, {a, e},{b, c}, {c, d}, {d, e}}

b) G= (V, E) and V ={a, b, c, d, e}and E={{a, b}, {b, c},{c, d}, {d, e}}

c) G= (V, E) and V ={a, b, c, d, e}and E={{a, b}, {a, c},{a, d}, {a, e}}

d) G= (V, E) and V ={a, b, c, d, e}and E={{a, b}, {a, c}, {d, e}}

e) G= (V, E) and V ={a, b, c, d, e}and E={{a, b}, {a, e},{b, c}, {c, d}}